DATABRICKS PRACTICAL ONLINE ASSESSMENT

Duration: 2 Hours | Total Marks: 100 | Mode: Practical Notebook-Based Assessment

Candidate Name		**Batch / Section**	
Register No.		**Date**	

General Instructions

- This assessment has one end-to-end practical case study. There are no MCQs.
- Use Databricks Notebook as the main working environment. PySpark and/or Spark SQL may be used.
- You may use concepts from Python, Pandas, NumPy, Spark DataFrames, SQL, Delta tables, and Databricks notebooks wherever suitable.
- Do not submit a Pandas-only solution. The main processing must demonstrate Databricks-based work.
- The solution is intentionally open-ended. Marks will be awarded for correctness, reliability, Databricks usage, and quality of reasoning.
- Write only the assumptions that are necessary for your solution. Do not over-explain theory.

Dataset

Use the Online Retail transaction dataset. It contains invoice-level retail transactions for a UK-based non-store online retail business.

CSV file:
https://raw.githubusercontent.com/databricks/Spark-The-Definitive-Guide/master/data/retail-data/all/online-retail-dataset.csv

Dataset reference: https://archive.ics.uci.edu/dataset/352/online+retail

Expected columns include InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country. Invoice numbers beginning with C indicate cancellations.

Case Study

A retail company wants to move from ad-hoc notebook analysis to a reliable Databricks-based reporting workflow. The management team wants a notebook that can identify trusted revenue, cancellation impact, customer/product patterns, and data-quality risks. The same workflow should also be able to handle a small late-arriving update batch without double-counting transactions.

Main Question

Build a Databricks solution that converts the raw retail transaction file into reliable business-ready outputs and demonstrates how the workflow handles a new incremental correction batch.

Assessment Requirements

Complete the following requirements in one Databricks notebook. You are free to choose the exact implementation approach, table structure, transformations, and output format, as long as the business requirements are satisfied.

Part	Requirement	Marks
A	Ingest the dataset into Databricks and prepare it for analysis. The notebook should show that the data was loaded correctly and that the schema, record count, and important columns were inspected.	15
B	Identify and handle data-quality risks that can affect reporting. Consider cancellations, negative or zero values, missing customer information, duplicate or repeated transaction patterns, date/time issues, and any other issues you find relevant.	20
C	Create reliable business metrics for the retail company. Include revenue-related metrics, cancellation/return impact, customer-level or country-level patterns, product-level performance, and time-based performance where suitable.	20
D	Create a small second batch manually using the same business structure. This batch must include new records, at least one correction to an existing invoice/order pattern, one cancellation-related case, and one additional new column. Process this batch safely so rerunning it does not inflate the output.	25
E	Produce final business-ready outputs that can be used for reporting. Your output should make it easy to understand revenue, quality issues, cancellation impact, and important business segments.	10
F	Notebook quality: clear execution order, readable code, meaningful table/view names, concise markdown headings, and reproducible logic.	10

Minimum Expected Final Outputs

- A cleaned or trusted transaction-level output suitable for analysis.
- A separate view/table/output that captures records not trusted for normal revenue reporting.
- At least three business summaries, chosen based on your own analysis direction.
- A compact before-and-after comparison showing the effect of the second batch.
- A final recommendation stating whether raw data should be used directly for dashboards or whether a controlled Databricks workflow is required.

Databricks Concepts Expected

The assessment is open-ended, but a strong answer should naturally demonstrate relevant Databricks concepts from the course. Use the right tool for the right part of the problem. Examples include Spark DataFrames, Spark SQL, temporary views, reusable tables, Delta features, update/insert-safe processing, schema handling, and notebook documentation.

Submission Format

- Submit the Databricks notebook or exported notebook file.
- The notebook must run from top to bottom after the dataset path is configured.
- Include only necessary markdown explanation. Do not write a long theory document.
- Do not submit screenshots only. Code and outputs must be present in the notebook.
- Mention any assumptions made while treating cancellations, negative quantities, missing customers, or late corrections.

Assessment Notes

- There is no single fixed solution. Different valid approaches can receive full marks.
- Marks are based on the quality of the Databricks workflow, correctness of metrics, handling of data issues, and ability to avoid duplicate or misleading reporting output.
- Hard-coded shortcuts, incomplete execution order, or output that works only for one displayed sample will lose marks.
- A solution that only prints a few aggregations without addressing data reliability will not be considered complete.

Candidate Reminder

Think like a data analyst working inside a production Databricks environment. The objective is not only to get numbers, but to make the numbers trustworthy.